By Theorem 13, columns 1, 2 and 4 are the basis for Col A.
∴ By defn, Dimension of Col A = 3.
By defn, Rank of A = 3.

Theorem 14: The Rank Theorem If a matrix A has n columns, then rank A + dim Nul A = n.

Theorem 15: The Basis Theorem

Let H be a p-dimensional subspace of $\mathbb{R}^n$. Any linearly independent set of exactly p elements in H is automatically a basis for H. Also, any set of p elements of H that spans H is automatically a basis for H.

Rank and the Invertible Matrix Theorem

Theorem: The Invertible Matrix Theorem (continued)

Let A be an nxn matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

m. The columns of A form a basis of $\mathbb{R}^n$.
n. Col A = $\mathbb{R}^n$
o. dim Col A = n
p. rank A = n
q. Nul A = {0}
r. dim Nul A = 0

Proof: By the equivalence of the statements (a), (e) and (h), we know that A is an invertible matrix if and only if the columns of A form a linearly independent set and spans $\mathbb{R}^n$. ∴ By Defn, A is an invertible matrix if and only if the columns of A form a basis of $\mathbb{R}^n$. ∴ (a) ⇔ (m).
By Defn, Col A = Span of columns of A. By the equivalence of statements (a) and (h), (a) ⇔ Col A = $\mathbb{R}^n$. ∴ (a) ⇔ (h).

~~Col A = $\mathbb{R}^n$ ⇔ dim Col A = n~~